

Last week : $SL(2)$ sector

$psu(2.2|4)$ Asymptotic Bethe Ansatz



+ Zhukovsky variable $x + \frac{1}{x} = u$

This week : (Nested) algebraic Bethe Ansatz

$su(2|2)$ asymptotic spin chain

§ Review of algebraic Bethe Ansatz

Want to diagonalize the Hamiltonian of $XXZ_{1/2}$ model:

$$H |s^1 \dots s^L\rangle = H_{b_1 \dots b_L}^{a_1 \dots a_L} |s^{b_1} \dots s^{b_L}\rangle, \quad s^a = \uparrow, \downarrow$$

Introduce auxiliary space, define monodromy

$$M_{b_1 \dots b_L, j}^{a_1 \dots a_L, i}(\lambda) = L_{b_L, i_L}^{a_L, i}(\lambda) \dots L_{b_1, j}^{a_1, i_2}(\lambda)$$

such that
$$H_{\vec{b}} = \frac{d}{d\lambda} \ln \text{tr} M_{\vec{b}}^{\vec{a}} \Big|_{\lambda = \frac{i}{2}}$$

Rewrite monodromy : $M^i_j = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}$

Bethe vacuum : $L_n(\lambda) \sim \begin{pmatrix} \lambda + i/2 & * \\ 0 & \lambda - i/2 \end{pmatrix}$

$$(A+D) |vac\rangle = \left(\lambda + \frac{i}{2}\right)^L + \left(\lambda - \frac{i}{2}\right)^L$$

Bethe state : $|u\rangle = B(u_1) \dots B(u_n) |vac\rangle$

Compute $(A+D) |u\rangle$ using RTT relations

$$A(\lambda) B(\mu) = \frac{\mu - \lambda + i}{\mu - \lambda} B(\mu) A(\lambda) + (\text{unwanted})$$

Nested Bethe Ansatz for $gl(N)$ spin chain

$$H|\vec{a}\rangle = H_{\vec{a},\vec{b}}|\vec{b}\rangle, \quad a_k \in \{1, 2, \dots, N\}$$

Want to find $T(\lambda) = \text{tr } M(\lambda)$ s.t.

$$T(\lambda) = \sum_n \left(\lambda - \frac{i}{2}\right)^n Q_n \rightsquigarrow H$$

Schematically,

$$M = \begin{pmatrix} A & B_1 & B_2 & \dots & B_{N-1} \\ C_1 & & & & \\ \vdots & & D & & \\ C_{N-1} & & & & \end{pmatrix}$$

Bethe vacuum : $T(\lambda) |vac\rangle = \Lambda_0(\lambda) |vac\rangle$

Level - I Bethe states :

$$|u\rangle = B^{c_1}(u_1) \cdots B^{c_n}(u_n) F_{c_1 \cdots c_n} |vac\rangle \quad \text{for } c_k = 1, 2, \dots, N-1$$

RTT relations : (usually $RTT \leftrightarrow \text{Yangian}$)

$$A(\lambda) B^c(\mu) = r^{cc'}(\lambda, \mu) B^{c'}(\mu) A(\lambda) + (\text{unwanted})$$

Eigenvalue equation :

$$T_{c_1 \cdots c_n}^{c'_1 \cdots c'_n}(\lambda) F_{c'_1 \cdots c'_n} = \Lambda(\lambda) F_{c_1 \cdots c_n}$$

→ Diagonalize $gl(N-1)$ indices by nested Bethe Ansatz

But this doesn't work in $N=4$ SYM beyond one-loop

$$S^a \leftrightarrow \mathcal{V} = \{ \phi, \psi, F, D\phi, D\psi, DF, \dots \}$$

Singleton representation of $psu(2,2|4)$

$$\text{non-compact spin chain} \leftrightarrow \mathcal{V}_1 \otimes \mathcal{V}_2 \otimes \dots \otimes \mathcal{V}_L$$

- Length-changing interaction: " $XYZ \leftrightarrow \psi\psi 1$ "
- Need to find R-matrix in x -variables
- Easy to guess rather than to prove

Beisert
0310252
0807.0799

§ Asymptotic Spin Chain

[Beisert, 0511082]

$SU(2|3)$ sector $\{z, \phi', \phi^2 \mid \psi', \psi^2\}$

Level-I vacuum, $|vac\rangle = |zz \dots z\rangle$

Level-I Bethe state (= level-II vacuum)

$$|z_1 \dots z_k\rangle = \sum_{n_1 < n_2 < \dots} e^{ip_1 n_1 + ip_2 n_2 + \dots} |z \dots \underbrace{\phi'}_{n_1} \dots \underbrace{\phi'}_{n_2} \dots z\rangle$$

Use coordinate Bethe Ansatz; don't know $\vec{B}(u)$

Level-II, III $\rightarrow$ same as nesting of Hubbard model

[Martins, Ramos, solv-int / 9712014]

To proceed, need more data:

- α -variable dependence
- S-matrix elements

Key idea: asymptotic chain $L \rightarrow \infty$ in $\mathbb{Z}^L$

- add or remove $\mathbb{Z}$ without changing state

$\leadsto$ central extension

Important because supersymmetry predicts

$\Delta(\lambda)$ of non-BPS states to all orders of λ

$\text{tr } \mathbb{Z}^L \rightarrow \text{BPS}$; $\text{tr } (\mathbb{Z}^\infty W^N) \rightarrow \text{"almost" BPS}$

Algebra $\mathfrak{psu}(2|2) \propto \mathbb{R}^3$

$$[R^a_b, J^c] = \delta_b^c J^a - \frac{1}{2} \delta_b^a \underline{J^c}$$

$$[L^\alpha_\rho, J^\gamma] = \delta^\gamma_\rho J^\alpha - \frac{1}{2} \delta^\alpha_\rho \underline{J^\gamma}$$

any generator

$$\{Q^\alpha_a, S^b_\rho\} = \delta_a^b L^\alpha_\rho + \delta_a^\rho R^b_a + \delta_a^b \delta_a^\rho \underline{C}$$

center

Central extension:

$$\{Q^\alpha_a, Q^\beta_b\} = \epsilon^{\alpha\beta} \epsilon_{ab} P$$

$$\{S^a_\alpha, S^b_\rho\} = \epsilon^{ab} \epsilon_{\alpha\rho} K$$

$psu(2|2) \propto \mathbb{R}^3$ is a limit of $d(2,1;\epsilon)$

Kac classification of simple super Lie alg.

$$d(2,1;\epsilon) = \text{Span} \left\{ \underbrace{R^a_b, L^x_\beta, M^A_B}_{\mathfrak{sl}(2) \oplus 3}, \underbrace{Q^{a\beta c}}_{\text{fermions}} \right\}$$

$$[M^A_B, J^C] = \delta^C_B J^A - (\text{trace})$$

$$[R^a_b, Q^{c\delta E}] = \delta^c_b Q^{a\delta E} - (\text{trace})$$

$$\begin{aligned} \{Q^{a\beta c}, Q^{d\epsilon F}\} = & \alpha \epsilon^{ak} \epsilon^{\beta\epsilon} \epsilon^{cF} R^a_k + \beta \epsilon^{ad} \epsilon^{\beta\epsilon} \epsilon^{cF} L^\epsilon_k \\ & + \gamma \epsilon^{ad} \epsilon^{\beta\epsilon} \epsilon^{cF} M^F_k \end{aligned}$$

Jacobi identity $\Rightarrow \alpha + \beta + \gamma = 0$

Rescaling $\Rightarrow \epsilon = \gamma/\alpha$

$$(Q^{\alpha\beta 1} \quad Q^{\alpha\beta 2}) = (\epsilon^{\alpha c} Q^{\beta}_c \quad \epsilon^{\beta\gamma} S^{\alpha}_{\gamma})$$

$$\begin{pmatrix} M^1_1 & M^1_2 \\ M^2_1 & M^2_2 \end{pmatrix} = \frac{1}{\epsilon} \begin{pmatrix} -c & p \\ -k & c \end{pmatrix}$$

Take the limit $\epsilon \rightarrow 0 \Rightarrow [M, M] = 0.$

$d(2,1;\epsilon) \rightsquigarrow$ Symmetry of $AdS_3 \times S^3 \times S^3 \times S^1$, $\epsilon \sim S^3$ radius ratio

Representations of $psu(2|2)$

$$Q_1^{a_{11}} Q_{12}^{a_{12}} Q_{21}^{a_{21}} Q_{22}^{a_{22}} |\phi\rangle \quad a_{i,j} \in \{0,1\}$$

$\Rightarrow$ dim. of minimum long rep is $2^4 = 16$

Minimum short rep. is $\{\phi^a | \psi^a\}$ for $(2|2)$

$$R^a_b |\phi^c\rangle = \delta^c_b |\phi^a\rangle - \frac{1}{2} S^a_b |\phi^c\rangle$$

as in $J^c = \phi^c$

$$L^a_\beta |\psi^\gamma\rangle = \delta^\gamma_\beta |\psi^a\rangle - (\text{trace})$$

We want to identify

$$\begin{pmatrix} \Phi^{11} & \Phi^{12} \\ \Phi^{21} & \Phi^{22} \end{pmatrix} \sim \underbrace{\phi^a \phi^b}_{\text{dim } \frac{1}{2}}, \quad \begin{pmatrix} D_{11} z & D_{12} z \\ D_{21} z & D_{22} z \end{pmatrix} \sim \underbrace{\psi^\alpha \psi^\beta}_{\text{dim } 1}$$

dim 1
dim 2

$$\underbrace{\{Q, Q^\dagger\}}_{\text{dim } \frac{1}{2}} \sim \underbrace{P}_{\text{dim } 1} \quad \underbrace{\{S, S^\dagger\}}_{\text{dim } -\frac{1}{2}} \sim \underbrace{K}_{\text{dim } -1}$$

$$\Rightarrow \begin{cases} Q^\alpha_a |\phi^b\rangle = a(x) \delta^b_a |\psi^\alpha\rangle & \text{dim} = 1 \\ Q^\alpha_a |\psi^\beta\rangle = b(x) \epsilon^{\alpha\beta} \epsilon_{ab} |\phi^b z^\dagger\rangle & \text{dim} = \frac{3}{2} \end{cases}$$

$\left\{ \begin{array}{l} \text{Closure of the action of } Q, S \text{ on } \phi, \psi \\ \text{Matching } su(2) \otimes su(2) \text{ indices} \\ \text{Matching dimensions} \end{array} \right.$

$\Rightarrow$ fixes the transformation rule up to coeff

$z^{\pm}$ means adding / removing z

$$\left\{ \begin{array}{l} S^a_{\alpha} |\phi^b\rangle = c(x) \epsilon^{ab} \epsilon_{\alpha\beta} |\psi^{\beta} z^{-}\rangle \\ S^a_{\alpha} |\psi^{\beta}\rangle = d(x) \delta^{\beta}_{\alpha} |\phi^a\rangle \end{array} \right.$$

Compute $\{Q, S\} = R + L + C$ on this basis:

$$\begin{aligned} Q^\alpha_a S^b_\beta \underline{|\phi^c\rangle} &= Q^\alpha_a c(x) \varepsilon^{bc} \varepsilon_{\beta\gamma} |\psi^\gamma z^-\rangle \\ &= b(x) c(x) \varepsilon^{bc} \varepsilon_{\beta\gamma} \varepsilon^{\alpha\gamma} \varepsilon_{ad} |\phi^d z^+ z^-\rangle \\ &= b(x) c(x) \delta^\alpha_\beta \left(\delta^b_a \underline{|\phi^c\rangle} - \delta^c_a \underline{|\phi^b\rangle} \right) \end{aligned}$$

$$\begin{aligned} \delta^b_\beta Q^\alpha_a \underline{|\phi^c\rangle} &= \delta^b_\beta a(x) \delta^c_a |\psi^\alpha\rangle \\ &= a(x) d(x) \delta^c_a \delta^\alpha_\beta \underline{|\phi^b\rangle} \end{aligned}$$

$$\delta^\alpha_\beta R^b_a \underline{|\phi^c\rangle} = \delta^\alpha_\beta \left(\delta^c_a \underline{|\phi^b\rangle} - \frac{1}{2} \delta^b_a \underline{|\phi^c\rangle} \right)$$

Compare coefficients of $\delta_\beta^\alpha \delta_a^b |\phi^c\rangle$ and $\delta_\beta^\alpha \delta_a^c |\phi^b\rangle$

$$|\phi^c\rangle: \quad b(x)c(x) = C - \frac{1}{2}$$

$$|\phi^b\rangle: \quad -b(x)c(x) + a(x)d(x) = 1$$

$$\text{Substitute back:} \quad C = \frac{ad-bc}{2} + bc = \frac{ad+bc}{2}$$

Action of $\{Q, S\} |\psi\rangle$ is the same due to automorphism

$$\phi \leftrightarrow \psi, \quad R \leftrightarrow L, \quad Q \leftrightarrow S$$

The other relations :

$$\{Q, Q\} \sim P \Rightarrow P|\chi\rangle = a(x)b(x) |\chi z^+\rangle$$

$$\{S, S\} \sim K \Rightarrow K|\chi\rangle = c(x)d(x) |\chi z^-\rangle$$

Recall that $P=K=0$ in the original $su(2|2)$

We impose $P=K=0$ on the total state :

$$|\chi_1, \chi_2, \dots\rangle = \sum_{n_1 \ll n_2 \ll \dots} e^{ip_1 n_1 + ip_2 n_2 + \dots} |\dots z \dots \underset{\hat{n}_1}{\chi_1} \dots \underset{\hat{n}_2}{\chi_2} \dots z\rangle$$

Then

$$P |x_1 \dots x_M\rangle = \sum_{k=1}^M \underline{a_k b_k} \prod_{l=k+1}^M e^{-i p_l} |x_1 \dots x_M z^+\rangle$$

rewrite $a_k b_k = \alpha (e^{-i p_k} - 1)$

Use:

$$(e^{-i p_k} - 1) e^{-i p_{k+1}} + (e^{-i p_{k+1}} - 1) = e^{-i(p_k + p_{k+1})} - 1$$

$$P |x_1 \dots x_M\rangle = \alpha \left(\prod_{k=1}^M e^{-i p_k} - 1 \right) |x_1 \dots x_M z^+\rangle$$

periodicity conditions, $P_{\text{tot}} \equiv 0 \pmod{2\pi}$

Transporting the shift operator $z^\pm$:

$$|z^\pm \chi_1, \chi_2, \dots\rangle = \sum_{n_1 \ll n_2} e^{ip_1 n_1 + ip_2 n_2 + \dots} | \dots \overset{\hat{n}_1}{\chi_1} \dots \overset{\hat{n}_2}{\chi_2} \dots \rangle$$

$$= \sum_{n_1 \ll n_2} e^{ip_1(n_1 \bar{f}_1) + ip_2 n_2 + \dots} | \dots \overset{\hat{n}_1}{\chi_1} \dots \overset{\hat{n}_2}{\chi_2} \dots \rangle$$

$$= e^{\bar{f} ip_1} |\chi_1, z^\pm, \chi_2, \dots\rangle$$

$$= e^{\bar{f} ip_1} |\chi_1, z^\pm \chi_2, \dots\rangle$$

$$= e^{\bar{f} ip_1 + \bar{f} ip_2} |\chi_1, \chi_2 z^\pm, \dots\rangle$$

Similarly

$$K |x_1 \dots x_n\rangle = \beta \left(\prod_{k=1} e^{i p_k} - 1 \right) |x_1 \dots x_n z^-\rangle$$

$$\text{if } c_k dk = \beta (e^{i p_k} - 1)$$

$$\text{In other words, } \begin{cases} P|x\rangle = \alpha (|z^+ x\rangle - |x z^+\rangle) \\ K|x\rangle = \beta (|z^- x\rangle - |x z^-\rangle) \end{cases}$$

Extra centers $\sim$ Gauge transform for adding/removing z

Use $\begin{cases} a_k b_k = \alpha (e^{-ip_k} - 1) \\ c_k d_k = \beta (e^{ip_k} - 1) \end{cases} \quad \& \quad a_k d_k - b_k c_k = 1$

We have $C = \sum_k C_k$

$$4C_k^2 = (a_k d_k + b_k c_k)^2 = 1 + 16 \alpha \beta \sin^2 \frac{p_k}{2}$$

Compare with $\gamma = \sum_k \left(\sqrt{1 + \frac{\lambda}{\pi^2} \sin^2 \frac{p_k}{2}} - 1 \right)$

gives $2C = 1 + \gamma, \quad 16 \alpha \beta = \lambda / \pi^2$

We also know that all-loop energy is written by

Zhukovsky variables:

$$E \sim \frac{i}{g} \left(x^+ - \frac{1}{x^+} - x^- + \frac{1}{x^-} \right)$$

$\Rightarrow a, b, c, d$ as a function of $x^\pm(p)$

$$(a, b, c, d) = \sqrt{\frac{g}{2}} \left(\gamma, \frac{-\alpha}{\gamma} \left(1 - \frac{x^-}{x^+} \right), \frac{i\gamma}{\alpha x^-}, \frac{x^+ - x^-}{i\gamma} \right)$$

$$\text{unitarity: } a = d^*, b = c^* \Rightarrow |\alpha| = 1, \gamma = \sqrt{i(x^- - x^+)}$$

Then

$$1 = ad - bc = \frac{g}{2i} \left(x^+ + \frac{1}{x^+} - x^- - \frac{1}{x^-} \right)$$

$$\text{solved if } x + \frac{1}{x} = u, \quad f^\pm(u) = f(u \pm \frac{i}{g})$$

Also

$$C = \frac{ad + bc}{2} = \frac{g}{2i} \left(x^+ - \frac{1}{x^+} - x^- + \frac{1}{x^-} \right)$$

$$\text{In general, } C = \sum_k \sqrt{1 + \underbrace{f(\lambda)}_{\text{fixed by localization / perturbative data}} \sin^2 \frac{p_k}{2}}$$

fixed by localization / perturbative data

$$\text{"g" depends on convention : } g^2 = \frac{N g_{YM}^2}{n \pi^2} \quad \text{with } n = 4, 8, 16$$

Crossing; relativistic case, $(E, p) = (m \cosh \theta, m \sinh \theta)$

$$\theta \rightarrow \theta + \pi i, (E, p) \rightarrow (-E, -p) \Rightarrow \text{anti-particle}$$

$x^\pm$ parameterize rapidity torus: $E > 0$ if $|x^\pm| \geq 1$

$$e^{ip} = \frac{x^+}{x^-}, \quad E = \frac{q}{i} \left(x^+ - \frac{1}{x^+} - x^- + \frac{1}{x^-} \right)$$

$$1 = \frac{q}{i} \left(x^+ + \frac{1}{x^+} - x^- - \frac{1}{x^-} \right)$$

$$x^\pm \rightarrow 1/x^\pm, (E, p) \rightarrow (-E, -p) \Rightarrow \text{anti-particle}$$

S -matrix : defined as the coefficients of
asymptotic wave-functions

$$|\psi\rangle \sim |\dots \chi_1^a \dots \chi_2^b \dots\rangle + S_{cd}^{ab}(p_1, p_2) |\dots \chi_2^c \dots \chi_1^d \dots\rangle$$

(+ when χ_1, χ_2 are nearby)

Then we constrain S by demanding

$$[J_1 + J_2, S_{12}] = 0 \quad \text{for all } J \text{ in } \mathfrak{psu}(2,2) \propto \mathbb{R}^3$$

$\Rightarrow$ automatically solves Yang-Baxter!

- Centrally-extended $\mathfrak{psu}(2|2) \ltimes \mathbb{R}^3$ is naturally
extended to Yangian algebra

- However, S-matrix for bound states need more data

- invariance under Yangian $[\hat{Y}_1 + \hat{Y}_2, S_{12}] = 0$

- unitarity, crossing, parity, CPT

- consistency with fusion

[Arutyunov, Frolov, 0803.4323]

also fixes "gauge choice" $(a, b, c, d) \rightarrow (\eta a, \frac{b}{2}, \eta c, \frac{d}{2})$

Example of computation:

$$\Delta(Q) = Q \otimes 1 + \underline{1 \otimes Q}$$

$\rightarrow e^{\bar{\psi} i P} \otimes Q$ in string theory convention

$$[\Delta(Q), S_{12}] |\phi_1^a \phi_2^b\rangle = 0$$

$$S_{12} |\phi_1^a \phi_2^b\rangle = A'_{12} |\phi_2^a \phi_1^b\rangle + B'_{12} |\phi_2^b \phi_1^a\rangle \\ + \frac{1}{2} C_{12} \epsilon^{ab} \epsilon_{\alpha\beta} |\psi_2^\alpha \psi_1^\beta\rangle \rightarrow$$

$$\Delta(Q) |\phi_1^a \phi_2^b\rangle = a_1 \delta_\alpha^a |\psi_1^\alpha \phi_2^b\rangle + a_2 \delta_\alpha^b |\phi_1^a \psi_2^\alpha\rangle$$

- Compute $S_{12} \Delta(Q) |\phi\phi\rangle$

$$S_{12} |\phi_1 \psi_2\rangle = G |\psi_2 \phi_1\rangle + H |\phi_2 \psi_1\rangle$$

$$S_{12} |\psi_1 \phi_2\rangle = L |\phi_2 \psi_1\rangle + K |\psi_2 \phi_1\rangle$$

Thus
$$S_{12} \Delta Q |\phi_1 \phi_2\rangle = (a_1 L + a_2 H) |\phi_2 \psi_1\rangle + (a_1 K + a_2 G) |\psi_2 \phi_1\rangle$$

- Compute $\Delta(Q) S_{12} |\phi\phi\rangle$

$$\Delta(Q) |\psi_2^\alpha \psi_1^\beta z^-\rangle = b_2 \varepsilon^{\gamma\alpha} \varepsilon_{cd} |\phi_2^d z^+ \psi_1^\beta z^-\rangle + b_1 \varepsilon^{\gamma\beta} \varepsilon_{cd} |\psi_2^\alpha \phi_1^d z^+ z^-\rangle$$

Use $z^+ z^- = 1$, $z^+ \chi_k = e^{-ipk} \chi_k z^+$

$$\Delta(Q) |\psi\psi z^- \rangle = b_2 e^{-ip_1} |\phi_2 \psi_1 \rangle + b_1 |\psi_2 \phi_1 \rangle$$

Also

$$(Q \otimes 1) \{ A' |\phi_2^a \phi_1^b \rangle + B' |\phi_2^b \phi_1^a \rangle + C |\psi\psi z^- \rangle \}$$

$$= a_2 \{ A' \delta_c^a |\psi_2^b \phi_1^b \rangle + B' \delta_c^b |\psi_2^a \phi_1^a \rangle + \dots \}$$

$$= a_2 A \{ \delta_c^a |\psi_2^b \phi_1^b \rangle + (a \leftrightarrow b) \}$$

$$+ a_2 B \{ \delta_c^a |\psi_2^b \phi_1^b \rangle - (a \leftrightarrow b) \} + \dots$$

if we use

$$\begin{cases} A' = A + B \\ B' = A - B \end{cases}$$

Compare coefficient of four terms:

$$\delta_c^a |\psi_2^\beta \phi_1^b\rangle, \delta_c^b |\psi_2^\beta \phi_1^a\rangle, \delta_c^a |\phi_2^b \psi_1^\beta\rangle, \delta_c^b |\phi_2^a \psi_1^\beta\rangle$$

$\Rightarrow$ equations for $\{A, B, C, G, H, K, L\}$ & $\{a_k, b_k\}$

Result: $A_{12} = S_{12}^0 \frac{x_1^+ - x_2^-}{x_1^- - x_2^+}$

$$\frac{A_{12} - B_{12}}{2} = S_{12}^0 \frac{x_1^+ - x_2^-}{x_1^- - x_2^+} \frac{x_2^+ - x_1^+}{x_2^+ - x_1^-} \frac{1 - 1/x_2^- x_1^+}{1 - 1/x_2^- x_1^-}$$

:

$$L_{12} = S_{12}^0 \frac{x_2^- - x_1^-}{x_2^- - x_1^+}$$

The S-matrix elements = Shastny's R-matrix of
1D Hubbard Model

Can diagonalize S-matrix using nested Bethe Ansatz

- coordinate BA [Beisert] [de Leeuw, 0705.2369]

- algebraic BA [Martins, Melo, 0703086]

$$|z \dots z\rangle \rightarrow |\phi^1 \dots \phi^1\rangle \rightarrow |\psi^{\alpha_1} \dots \psi^{\alpha_k}\rangle, \quad \phi^2 \sim \psi^1 \psi^2$$

($\alpha = 1, 2$)

SU(2) symmetry

Composite
excitation

Su(2|3) Bethe eq: $(v_k = y_k + 1/y_k)$

$$\left(\frac{z_k^+}{z_k^-} \right)^{K_0} = \prod_{l \neq k}^{K_1} S^0(x_l, x_k) \frac{z_k^+ - z_l^-}{z_k^- - z_l^+} \prod_{l=1}^{K_2} \frac{z_k^- - y_l}{z_k^+ - y_l}$$

$$1 = \prod_{l=1}^{K_1} \frac{y_k - x_l^+}{y_k - x_l^-} \prod_{l=1}^{K_3} \frac{v_k - w_l + i/g}{v_k - w_l - i/g}$$

$$1 = \prod_l^{K_2} \frac{w_k - v_l + i/g}{w_k - v_l - i/g} \prod_{l \neq k}^{K_3} \frac{w_k - w_l - 2i/g}{w_k - w_l + 2i/g}$$



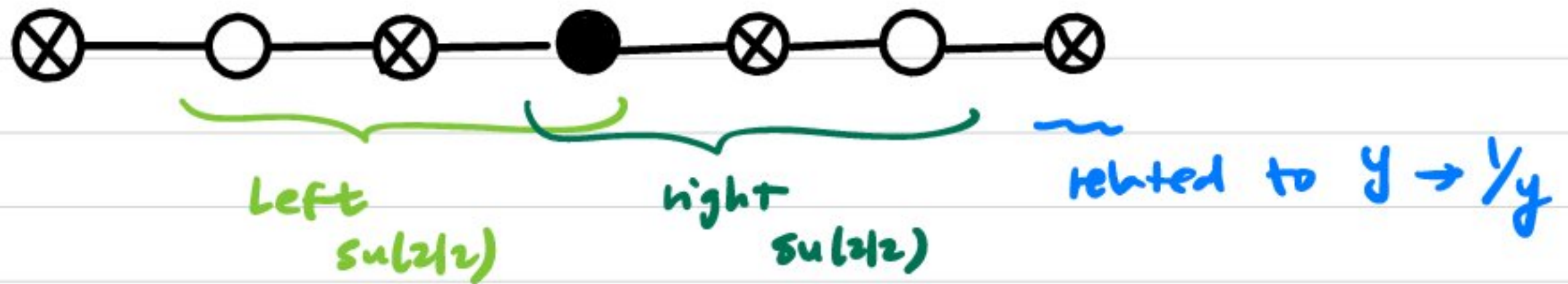
Same as XXXY₂

To get BAE for $N=4$ SYM,

we need to glue two wings of $su(2|2)$

$$PSU(2,2|4) \longrightarrow (PSU(2|2) \ltimes \mathbb{R}^3)^2$$

by identifying center C (or momentum-carrying node)



Can derive similar eqs from $SU(1|1)^2$ for AdS_3

but one copy of $SU(1|1)$ for AdS_2 is less strong

Symmetry doesn't fix the scalar $S_0(x_1, x_2)$

$\leadsto$ dressing factor

We need crossing relations

- if we assume "effective 2-dim QFT" describing asymptotic ch.m., we should have "crossing"
- or we impose that S-matrix scatters trivially against singlet = (particle, anti-particle composite)

